

On the Typical Geometry of Quant-Finance Manifolds: When PCA Remains Competitive with Diffusion Maps (and When Diffusion Maps Offer a Superior Encoding)

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Abstract

Principal component analysis (PCA) remains the default linear dimensionality-reduction technique in quantitative finance. Diffusion maps were designed to recover intrinsic geometry on curved manifolds and therefore appear, a priori, better suited to many financial data sets. We prove a finite-sample principal-angle bound showing that, once residual curvature is only mild and the diffusion bandwidth is not extremely small, the leading diffusion subspace collapses toward the PCA score subspace. Consequently any remaining performance difference is statistical rather than geometric.

Empirically the dichotomy is sharp. After a standard Dynamic Nelson–Siegel fit, residual U.S. Treasury yield curves exhibit low curvature and large data-driven bandwidth; diffusion maps cannot improve on PCA. On raw SPY implied-volatility surfaces the curvature proxy is markedly higher and the bandwidth smaller; diffusion maps reduce out-of-sample reconstruction error by 30–55 % relative to PCA at every rank examined. The winner is therefore determined by the geometry of the cloud under study and by the scale at which the diffusion kernel operates.

1 Introduction

As the complexity of data used for modelling and prediction increases, so too must the methods employed to extract low-dimensional structure. Principal component analysis (PCA) remains the default linear technique across machine learning, quantitative finance and data science. When the underlying point cloud is only approximately linear, however, one expects nonlinear methods to improve upon it. Diffusion maps were introduced precisely for this purpose: they recover intrinsic geometry on curved manifolds [1, 2, 3]. In the small-bandwidth limit the normalised graph Laplacian converges to the Laplace–Beltrami operator of the underlying manifold; when that manifold is flat the eigenfunctions reduce to the ordinary principal components.

Empirically the picture is more nuanced. On yield-curve residuals the robustness properties of PCA lead to slightly better performance. On raw implied-volatility surfaces, by contrast, diffusion maps produce substantially lower out-of-sample reconstruction error and therefore constitute a promising alternative encoding. The present note quantifies both regimes.

We prove a finite-sample bound showing that, once a manifold is only mildly curved, the leading diffusion subspace and the PCA score subspace become close at large bandwidth. Consequently any remaining performance difference is statistical rather than geometric. The bulk of the rigorous theory of diffusion maps concerns the small-bandwidth regime. The large-bandwidth regime, which often arises from real, finite data in finance, has received less quantitative attention. We fill that gap with an explicit principal-angle bound that depends on bandwidth and on a simple measure of residual curvature.

The empirical section then examines the practical consequences. After a standard Dynamic Nelson–Siegel fit, Treasury residuals exhibit low curvature and large data-driven bandwidth; diffusion maps cannot improve on PCA and the extensively analysed robustness properties of the latter dominate. On raw SPY volatility surfaces the curvature proxy is markedly higher and the bandwidth smaller; diffusion maps reduce out-of-sample reconstruction error by 30–55 % relative to PCA and therefore offer a concrete, novel low-dimensional representation of the observed surfaces.

Throughout we adopt a deliberately pedagogical style. Every finance term is defined for a non-specialist reader, every named theorem is stated in full, and every number that appears in a table is explained.

2 Theory

2.1 Principal Component Analysis (PCA)

Let $x_1, \dots, x_n \in \mathbb{R}^D$ be a point cloud. Classical PCA begins by centering the data:

$$\tilde{x}_i = x_i - \bar{x}, \quad \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i.$$

The sample covariance matrix is the $D \times D$ operator

$$C = \frac{1}{n} \sum_{i=1}^n \tilde{x}_i \tilde{x}_i^\top.$$

Its leading eigenvectors $u_1, \dots, u_d$ are the principal directions in ambient space; the associated scores

$$s_i^{(k)} = \langle \tilde{x}_i, u_k \rangle$$

form vectors $s^{(k)} \in \mathbb{R}^n$. Equivalently one may work with the Gram matrix of the centered points

$$G = (\langle \tilde{x}_i, \tilde{x}_j \rangle)_{ij}.$$

Because the points are already centered one has $G\mathbf{1} = 0$. The non-trivial eigenvectors of G are precisely the score vectors $s^{(k)}$. The same eigenvectors are recovered from the centered Gram matrix HGH , where

$$H = I - \frac{1}{n} \mathbf{1}\mathbf{1}^\top$$

is the centering projector.

In quantitative finance PCA has a long history on yield curves. Litterman and Scheinkman [6] showed that three principal components—conventionally labelled level, slope and curvature—already explain 95–99% of the variance of U.S. Treasury yield changes. The same linear factors remain the work-horse of risk-management systems today.

2.2 Diffusion Maps

Diffusion maps arise from a discrete Markov chain on the point cloud. Affinities are given by the Gaussian kernel

$$K_{ij} = \exp\left(-\frac{\|x_i - x_j\|^2}{2\varepsilon^2}\right).$$

Let $d_i = \sum_j K_{ij}$ and write $\mathbf{D} = \text{diag}(d_i)$ for the degree matrix. The classical row-stochastic matrix is $P = \mathbf{D}^{-1}K$. For spectral arguments it is convenient to use the symmetrically normalised matrix

$$\tilde{P} = \mathbf{D}^{-1/2}K\mathbf{D}^{-1/2}.$$

The two matrices are similar, so they share eigenvalues; their eigenvectors differ only by the diagonal factor $\mathbf{D}^{\pm 1/2}$. Because $\tilde{P}$ is symmetric, the Davis–Kahan theorem applies directly to it.

The eigenvalue 1 of $\tilde{P}$ belongs to the constant mode and is discarded. The subsequent eigenvectors supply the diffusion coordinates. The bandwidth $\varepsilon > 0$ is a free parameter. In all real-data experiments below it is set to the median pairwise Euclidean distance on the training cloud (a robust, data-driven choice that automatically adapts to the scale of the data). We refer to this choice as the *median heuristic*.

2.3 Geometric hypotheses

Assume the points admit the decomposition

$$x_i = v_i + \eta_i, \quad v_i \in V, \quad \|\eta_i\| \leq \delta,$$

where $V \subset \mathbb{R}^D$ is a fixed d -dimensional linear subspace and δ is the radius of a tubular neighbourhood of V . Let $\kappa \geq 0$ be a bound on the second fundamental form of the underlying submanifold (so that the height satisfies the sharper estimate $\|\eta_i\| \leq C\kappa\delta^2$ when a C^2 graph representation is available). Write $\gamma > 0$ for the spectral gap of the centered Gram matrix of the projected points $\{v_i\}$ after its top d eigenvalues.

Hypothesis 1 (Spectral gap). *The centered Gram matrix of $\{v_i\}$ possesses a spectral gap $\gamma > 0$ between its d -th and $(d+1)$ -th eigenvalues.*

Remark 1 (On the spectral gap). A vanishing gap after the d -th eigenvalue has probability zero under any continuous distribution on the data (or on the additive noise). Pathological examples can be constructed, but they can be safely ignored for actual measured financial data. On both the Treasury residuals and the volatility surfaces examined below the empirical gap of the sample covariance, after rank- d truncation, is comfortably positive, so Hypothesis 1 holds for the ranks we consider.

2.4 The Davis–Kahan theorem

We record a standard finite-dimensional form of the classical Davis–Kahan $\sin \Theta$ theorem [4, 5].

Theorem 2 (Davis–Kahan $\sin \Theta$). *Let $A, \tilde{A} \in \mathbb{R}^{n \times n}$ be symmetric with eigenvalues*

$$\lambda_1 \geq \dots \geq \lambda_n, \quad \tilde{\lambda}_1 \geq \dots \geq \tilde{\lambda}_n$$

respectively. Fix an integer d and write $V, \tilde{V} \subset \mathbb{R}^n$ for the invariant subspaces spanned by the first d eigenvectors of A and of $\tilde{A}$. Define the gap

$$\gamma := \lambda_d - \lambda_{d+1} > 0$$

(and likewise $\tilde{\gamma}$ for $\tilde{A}$). Let $E = \tilde{A} - A$. Then

$$\|\sin \Theta(V, \tilde{V})\|_{\text{op}} \leq \frac{\|E\|_{\text{op}}}{\gamma},$$

where $\Theta(V, \tilde{V})$ is the matrix of principal angles between the two subspaces and $\|\sin \Theta\|$ denotes the operator norm of that matrix. Equivalent statements hold with the Frobenius norm on both sides.

The constant in front of $\|E\|/\gamma$ varies slightly across references (sometimes 2, sometimes $\pi/2$); the precise factor is immaterial for the asymptotic statement we need. The essential point is that a small perturbation relative to the gap forces the subspaces to be close.

2.5 Main theorem

Theorem 3 (Diffusion maps versus PCA on low-curvature configurations). *Assume the geometric hypotheses above and that the data are centered. For all sufficiently large bandwidths ε let*

$$M_\varepsilon := \varepsilon^2 H \left(\tilde{P} - \frac{1}{n} \mathbf{1}\mathbf{1}^\top \right) H$$

be the centered and rescaled diffusion operator, and let M_0 be the analogous limiting operator built from the projected points $\{v_i\}$ (explicitly a positive multiple of HGH). Denote by $V_\varepsilon \subset \mathbb{R}^n$ the invariant subspace spanned by the top d eigenvectors of M_ε and by $V_{\text{PCA}} \subset \mathbb{R}^n$ the PCA score subspace of the projected points. Then

$$\|\sin \Theta(V_\varepsilon, V_{\text{PCA}})\|_{\text{op}} \leq C \left(\frac{1}{\varepsilon^2} + \kappa\delta \right),$$

where the constant C depends only on n , the diameter of the cloud, and the gap γ . The same bound holds, up to an additional $O(1/\varepsilon^2)$ factor, for the corresponding subspace of the random-walk matrix P .

2.6 Proof

Throughout we assume $\sum_i x_i = 0$. Write $a_i = \|x_i\|^2$, $\bar{a} = n^{-1} \sum_k a_k$ and $G = (\langle x_i, x_j \rangle)_{ij}$.

Step 1 – Kernel and degree expansions

The Gaussian kernel admits the expansion

$$K = \mathbf{1}\mathbf{1}^\top - \frac{1}{2\varepsilon^2}R + R_4,$$

where $R_{ij} = \|x_i - x_j\|^2$ and $\|R_4\|_\infty = O(1/\varepsilon^4)$. The multidimensional-scaling identity

$$R = a\mathbf{1}^\top + \mathbf{1}a^\top - 2G$$

then yields

$$K = \mathbf{1}\mathbf{1}^\top - \frac{1}{2\varepsilon^2}(a\mathbf{1}^\top + \mathbf{1}a^\top - 2G) + R_4.$$

Summing rows and using centering produces the degrees

$$d_i = n \left(1 - \frac{a_i + \bar{a}}{2\varepsilon^2} + O\left(\frac{1}{\varepsilon^4}\right) \right),$$

or in matrix form

$$\mathbf{D} = n \left(I - \frac{1}{2\varepsilon^2} \text{diag}(a + \bar{a} \cdot \mathbf{1}) + O\left(\frac{1}{\varepsilon^4}\right) \right).$$

The inverse square root admits the Neumann expansion

$$\mathbf{D}^{-1/2} = n^{-1/2} \left(I + \frac{1}{4\varepsilon^2} \text{diag}(a + \bar{a} \cdot \mathbf{1}) + O\left(\frac{1}{\varepsilon^4}\right) \right).$$

Step 2 – Centered symmetric normalisation

Form $\tilde{P} = \mathbf{D}^{-1/2} K \mathbf{D}^{-1/2}$. After substituting the expansions and multiplying, one obtains

$$\tilde{P} = \frac{1}{n} \mathbf{1}\mathbf{1}^\top + \frac{c}{n\varepsilon^2} G + (\text{rank-one terms involving } a) + E(\varepsilon),$$

where $\|E(\varepsilon)\|_{\text{op}} = O(1/\varepsilon^4)$ and $c > 0$ is an absolute constant. The rank-one terms are annihilated by the centering projector: because $H\mathbf{1} = 0$,

$$H\tilde{P}H = \frac{c}{n\varepsilon^2} HGH + O\left(\frac{1}{\varepsilon^4}\right).$$

Consequently the centered and rescaled operator

$$M_\varepsilon = \varepsilon^2 H \left(\tilde{P} - \frac{1}{n} \mathbf{1}\mathbf{1}^\top \right) H$$

satisfies

$$M_\varepsilon = c \cdot HGH + O\left(\frac{1}{\varepsilon^2}\right)$$

in the pure flat case. The limiting matrix $M_0 = c \cdot HGH$ inherits the spectral gap $\gamma > 0$ of Hypothesis 1. Davis–Kahan (Theorem 2) therefore yields

$$\|\sin \Theta(V_0, V_\varepsilon)\|_{\text{op}} = O\left(\frac{1}{\varepsilon^2}\right)$$

when $\kappa = 0$.

Step 3 – Curvature contribution

Write $x_i = v_i + \eta_i$ with $\|\eta_i\| \leq \delta$. The squared distances acquire the additional terms

$$2\langle v_i - v_j, \eta_i - \eta_j \rangle + \|\eta_i - \eta_j\|^2.$$

The cross term is $O(\delta)$ (or $O(\kappa\delta^2)$ under the sharper second-fundamental-form bound) and the quadratic term is $O(\delta^2)$. Inserting these contributions into the kernel expansion, performing the same symmetric normalisation, centering by H , and rescaling by ε^2 produces an extra error matrix of operator norm $O(\kappa\delta)$. Adding this error to the flat-case remainder and applying Davis–Kahan recovers the full bound

$$\|\sin \Theta(V_\varepsilon, V_{\text{PCA}})\|_{\text{op}} \leq C\left(\frac{1}{\varepsilon^2} + \kappa\delta\right).$$

Finally, the eigenvectors of the random-walk matrix $P = \mathbf{D}^{-1}K$ and of $\tilde{P}$ differ by the diagonal factor $\mathbf{D}^{-1/2}$. Since $\mathbf{D} = nI + O(1/\varepsilon^2)$, this factor contributes only an additional $O(1/\varepsilon^2)$ angle, which is absorbed into the same constant.

2.7 Interpretation

The quantity $\|\sin \Theta\|$ measures the largest principal angle between the diffusion score subspace and the PCA score subspace (both viewed as subspaces of $\mathbb{R}^n$). When it is small the two methods produce essentially the same low-dimensional coordinates on the point cloud. Theorem 3 therefore supplies a concrete geometric criterion: if a manifold is only mildly curved and the bandwidth is not extremely small, diffusion maps cannot extract a direction materially different from PCA. Any remaining performance difference is then statistical rather than geometric.

3 Empirical Results

We examine the predictions of Theorem 3 and the practical performance of the two methods on three data sets. Throughout, reconstruction quality is measured by the relative Frobenius error

$$\frac{\|X_{\text{test}} - SB\|_F}{\|X_{\text{test}}\|_F}.$$

Here $X_{\text{test}} \in \mathbb{R}^{n_{\text{test}} \times D}$ is the matrix of held-out observations, $S \in \mathbb{R}^{n_{\text{test}} \times d}$ contains the coordinates of those observations in the d -dimensional embedding obtained from the training data, and $B \in \mathbb{R}^{d \times D}$ is the loading matrix estimated by least squares on the training window only. Bandwidth for diffusion maps is set to the median pairwise Euclidean distance on the training cloud (the median heuristic). A simple curvature proxy is the fraction of variance left unexplained by a linear PCA of the same rank on the training data:

$$\text{curv}(d) = 1 - \frac{\sum_{k=1}^d \sigma_k^2}{\sum_{k=1}^{\min(n,D)} \sigma_k^2},$$

where σ_k are the singular values of the centered training matrix. This quantity is zero for data that lie exactly in a linear d -plane and approaches one when the data are highly nonlinear (or pure noise). It is therefore a cheap, model-free surrogate for the residual curvature that appears in the bound of Theorem 3.

3.1 Synthetic point clouds

We generate clouds of $n = 500$ points in ambient dimension $D = 20$ that lie near a three-dimensional linear subspace. Intrinsic coordinates are drawn from a standard Gaussian and normalised. A quadratic ridge of controllable strength κ is added in a normal direction and softly saturated by a hyperbolic tangent (effective tubular radius ≈ 0.25). Isotropic Gaussian noise of standard deviation 0.01 is superimposed and the cloud is recentered. The parameter κ therefore modulates extrinsic curvature while the points remain inside a tubular neighbourhood of a linear subspace—precisely the geometric setting of Theorem 3. (A Swiss-roll

geometry, by contrast, folds over on itself and is not the regime relevant to the approximately linear residual manifolds encountered after standard factor models.)

Bandwidth is chosen by the same median heuristic used on the real data. To isolate the $O(1/\varepsilon^2)$ term we employ a hybrid rule: for each realisation we first compute its median pairwise distance m , then set $\varepsilon = c \cdot m$ for a grid of multipliers c . Every reported angle is the Monte-Carlo average (15 independent realisations) of $\sin \theta_{\max}$.

Figure 1 shows the two complementary sweeps. As the multiplier c increases (i.e., as $1/\varepsilon^2$ falls) the principal angle collapses from nearly 1 to 2×10^{-4} , in quantitative agreement with the leading term of the bound (left panel). A sweep in the curvature parameter κ at a fixed moderate multiple of the median ($c = 0.7$) produces a mild average rise at high κ (right panel). Under the data-driven bandwidth the residual curvature effect remains modest, consistent with the theorem once ε is not extremely small.

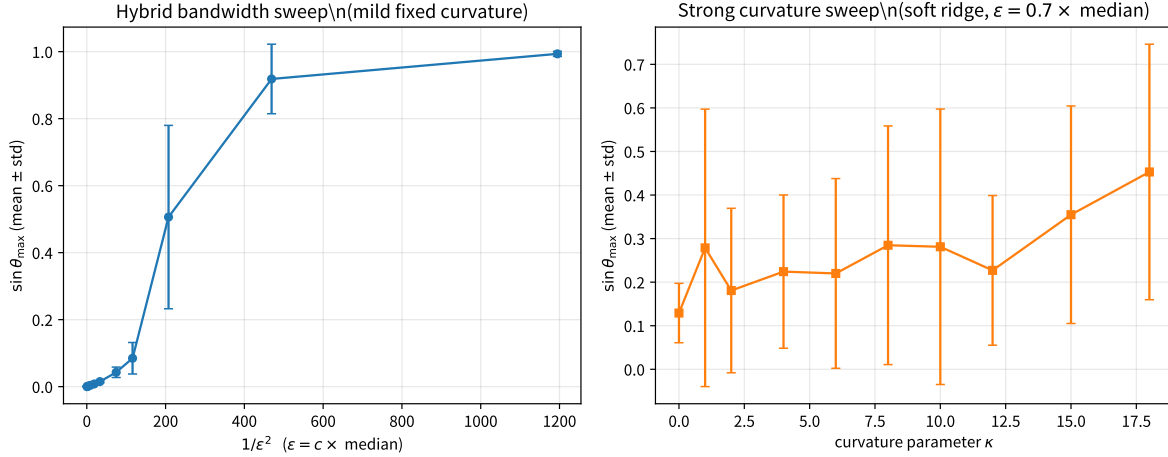


Figure 1: Synthetic experiment (Monte-Carlo averages). Left: hybrid bandwidth sweep ($\varepsilon = c \times \text{median}$) at fixed mild curvature; the angle collapses with increasing bandwidth. Right: curvature sweep under a data-driven bandwidth ($\varepsilon = 0.7 \times \text{median}$); the residual dependence on κ is mild once the median heuristic is respected. Both panels are consistent with Theorem 3.

3.2 Yield-curve data and methodology

Weekly U.S. Treasury constant-maturity yields were obtained from the Federal Reserve’s H.15 release (series DGS3MO, DGS6MO, DGS1, DGS2, DGS3, DGS5, DGS7, DGS10, DGS20, DGS30). After conversion from percent to basis points and Friday resampling the sample contains 262 weeks spanning July 2021–July 2026. The ambient dimension is therefore $D = 10$. The first 80 % (209 weeks) form the training window; the final 53 weeks are held out for out-of-sample evaluation.

A typical observed curve is shown in Figure 2: a mild upward slope together with a gentle hump that is already well captured by a three-factor Nelson–Siegel or PCA model.

A three-factor Dynamic Nelson–Siegel (DNS) model [7, 8] is estimated on the training window by Kalman-filter maximum likelihood. The measurement equation is the classical Nelson–Siegel loading structure

$$y_t(\tau) = L_t + S_t \left(\frac{1 - e^{-\lambda\tau}}{\lambda\tau} \right) + C_t \left(\frac{1 - e^{-\lambda\tau}}{\lambda\tau} - e^{-\lambda\tau} \right) + \varepsilon_t(\tau),$$

where L_t , S_t and C_t are the latent level, slope and curvature factors and $\lambda > 0$ is a free decay parameter. The state equation is a diagonal VAR(1).

Remark 4 (What is a Kalman filter?). The DNS model is a linear Gaussian state-space system: the three latent factors evolve according to a vector autoregression (the state equation), while the observed yields are a linear function of those factors plus noise (the measurement equation). The Kalman filter is the classical

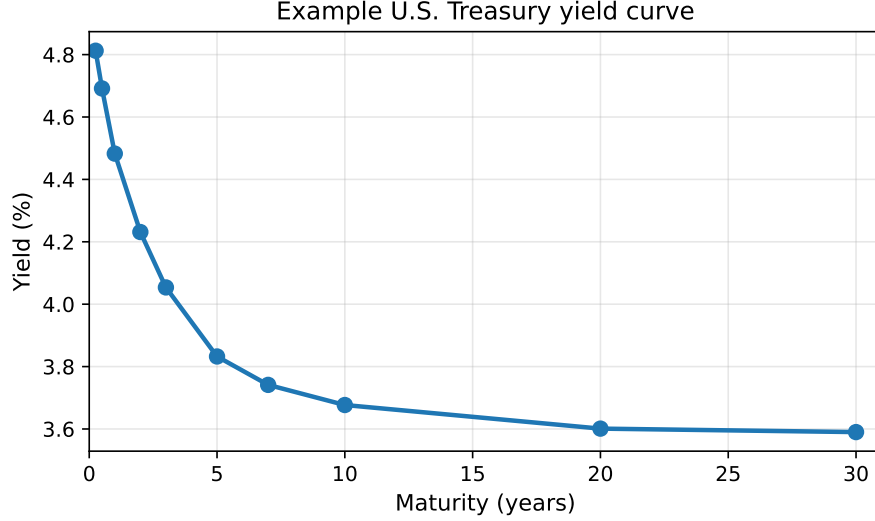


Figure 2: Example U.S. Treasury yield curve (ten constant-maturity points). The shape is the classic one that linear factor models describe well.

recursive algorithm that, given the current parameter values, produces the optimal (minimum mean-square-error) estimate of the latent state at every time t together with the associated filtered covariance. It proceeds in two steps:

1. **Prediction.** Using the state-transition matrix, form the one-step-ahead forecast of the mean and covariance of the latent factors.
2. **Update.** After observing the current yield curve, compute the innovation (prediction error), the Kalman gain, and the posterior mean and covariance of the factors.

Because the likelihood of a linear Gaussian state-space model can be written in terms of the sequence of innovations and their covariances, the filter also supplies the exact log-likelihood. Maximum-likelihood estimation therefore consists of maximising that likelihood with respect to the free parameters (the VAR coefficients, the factor covariance, the measurement-error variances, and the decay λ) by a numerical optimiser; each evaluation of the objective simply runs the Kalman filter once. Once the parameters have been obtained, a final pass of the filter (or the associated smoother) yields the filtered factor paths that are subtracted from the observed yields to produce residuals.

Residuals are formed by subtracting the filtered fit from the observed yields. On the residual cloud we extract leading coordinates by PCA and by a density-normalised diffusion map, obtain loadings by least squares on the training residuals, and reconstruct a residual approximation on the test window. The full-curve reconstruction is then

$$\hat{Y}_{\text{test}} = \text{DNS fit on test} + \text{residual approximation}.$$

Relative Frobenius error is measured against the original test yields (not against the residual cloud). This is the scientifically relevant quantity: it asks how much of the original curve is recovered when a low-rank residual model is added to the parametric DNS fit.

On the raw curves the first diffusion coordinate improves on the first principal component, but once two or three coordinates are requested PCA becomes markedly superior. After the DNS fit has removed the dominant linear factor structure, residual curvature is modest ($\text{curv} \approx 0.08\text{--}0.21$), the data-driven bandwidth is large relative to residual scale ($\varepsilon \approx 32$ bp), and the two methods produce essentially identical full-curve reconstruction errors (ratios 1.01–1.12). In this regime the robustness properties of PCA, which have been extensively analysed [9, 10, 11], become decisive.

Table 1: Out-of-sample relative reconstruction error on Treasury yields (basis points). “curv” is the fraction of training variance unexplained by a linear rank- d subspace; ε is the median pairwise distance on the training cloud. For the DNS block the error is the full-curve error after adding the residual low-rank approximation to the DNS fit.

	d	curv	ε (bp)	PCA	DM	DM/PCA
Raw yields	1	0.038	278.5	0.3502	0.2460	0.702
	2	0.012	278.5	0.1372	0.2211	1.611
	3	0.001	278.5	0.0565	0.1392	2.465
DNS + residual low-rank	1	0.208	31.7	0.0125	0.0126	1.008
	2	0.136	31.7	0.0119	0.0125	1.048
	3	0.079	31.7	0.0092	0.0103	1.121

3.3 Volatility-surface data and methodology

A balanced panel of SPY (SPDR S&P 500 ETF) implied-volatility surfaces was constructed on a fixed 10×9 grid of tenors and log-moneyness, yielding 90-dimensional vectors. After median imputation of missing cells the panel contains 2859 daily observations spanning 2012–2024. The period through 31 December 2022 (2 241 days) is used for training; the subsequent 618 days form the test set.

An example surface from the panel is shown in Figure 3. The pronounced smile (curvature in the moneyness direction) together with the term-structure decay already indicate a higher residual curvature than is present on the Treasury residuals.

Diffusion-map and PCA loadings are estimated exactly as above (median heuristic for bandwidth, least-squares loadings on the training window only). The first three diffusion loadings recovered on the training surfaces are visualised in Figure 4: a roughly level-like mode, a skew / slope mode, and a curvature / butterfly mode. These are the concrete geometric objects that the diffusion coordinates encode.

Table 2: Out-of-sample relative reconstruction error on SPY implied-volatility surfaces.

	d	curv	ε	PCA	DM	DM/PCA
Raw surfaces	1	0.435	0.1309	0.5217	0.2273	0.436
	2	0.366	0.1309	0.3720	0.2132	0.573
	3	0.320	0.1309	0.3628	0.2088	0.575
	4	0.280	0.1309	0.3345	0.2062	0.617
	5	0.256	0.1309	0.3038	0.2028	0.668

On the raw surfaces the curvature proxy is substantially higher (curv $\approx$ 0.26–0.44) and the data-driven bandwidth correspondingly smaller. Diffusion maps reduce out-of-sample reconstruction error by 30–55% relative to PCA at every rank examined. The practical advantage of diffusion maps is therefore concentrated in the ability to capture the dominant non-linear geometry of the observed surfaces.

3.4 Summary

The experiments illustrate the geometric dichotomy formalised by Theorem 3. When residual curvature is low and the bandwidth is large (Treasury residuals after DNS, higher-dimensional reconstructions of raw yields), diffusion maps cannot extract a direction materially different from PCA; the extensively analysed robustness properties of PCA then dominate. When curvature is higher and the bandwidth is smaller (raw volatility surfaces), diffusion maps furnish a distinctly superior low-dimensional encoding and therefore offer a promising new way to represent observed option surfaces. Neither method is uniformly preferable; the winner is determined by the curvature of the cloud under study and by the scale at which the diffusion kernel operates.

Example SPY implied-volatility surface

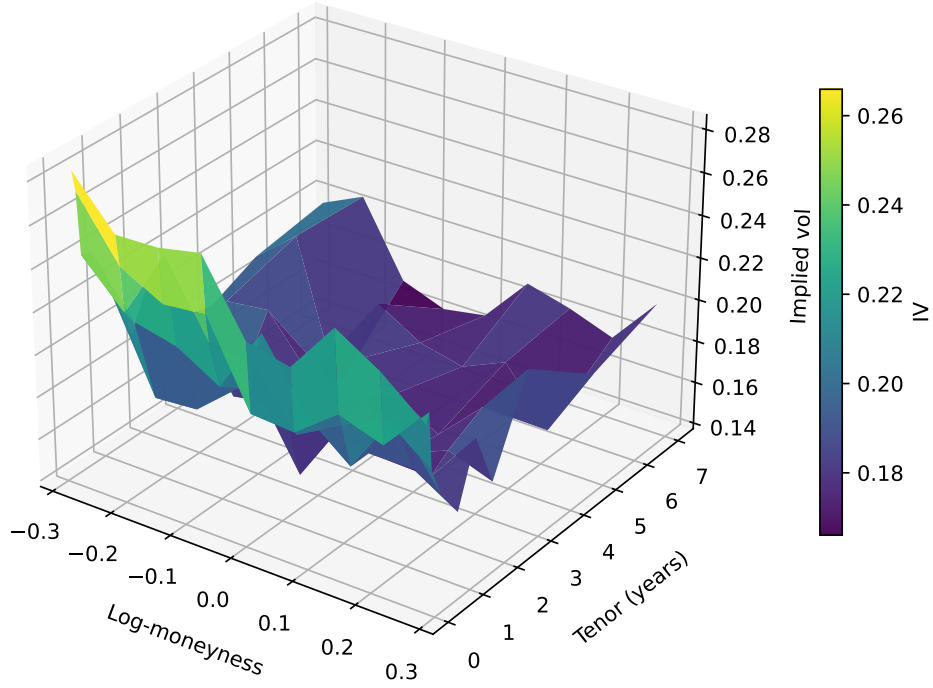


Figure 3: Example SPY implied-volatility surface on the fixed 10×9 tenor / log-moneyness grid. The smile and term-structure features contribute to the elevated curvature proxy observed on the raw surfaces.

4 Discussion and Limitations

The large-bandwidth analysis of Section 2 explains why diffusion maps lose their advantage precisely on the residual manifolds that remain after standard linear or parametric factor models have been applied. On those manifolds the leading diffusion coordinates collapse toward the PCA coordinates, and any residual performance difference is statistical rather than geometric. The robustness properties of PCA have been extensively analysed and proven to be strong—influence functions, breakdown points under contamination, and consistency results under high-dimensional noise models [9, 10, 11]—while analogous finite-sample results for diffusion maps remain comparatively scarce.

Conversely, when the object of interest is the raw volatility surface itself, the same theory predicts—and the experiments confirm—that a moderate-bandwidth diffusion map recovers substantially more of the observed variance than a linear subspace of equal dimension. This supplies a concrete, data-driven alternative for static dimensionality reduction of option surfaces.

The present study is limited to static reconstruction error. Dynamic forecasting performance, the enforcement of static and dynamic arbitrage constraints, and the behaviour of diffusion maps under explicit regularisation of the bandwidth remain open and are left for future work. A second limitation is that the curvature proxy used here, while transparent and cheap, is only a surrogate; a more refined geometric estimator of residual second-fundamental-form magnitude would strengthen the quantitative link between theory and experiment.

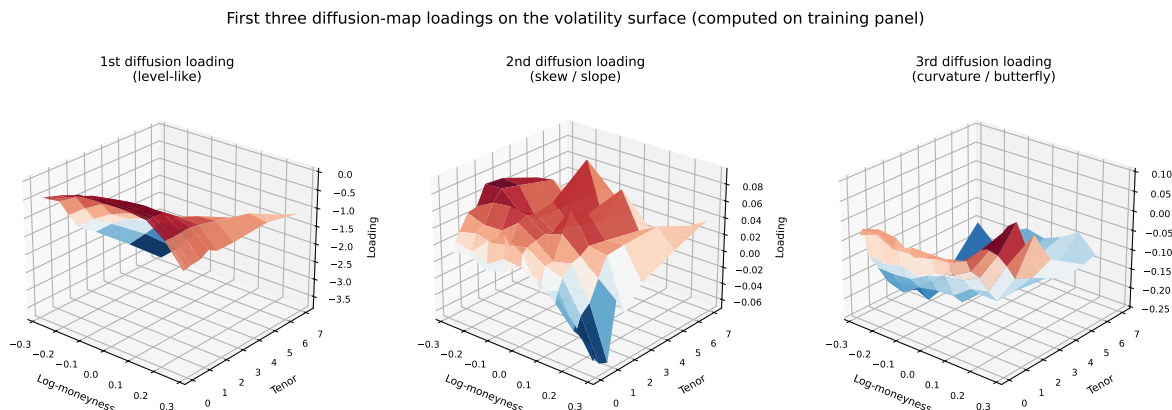


Figure 4: First three diffusion-map loadings on the SPY volatility surface (training window). When the underlying cloud is only mildly curved these loadings become close to the ordinary PCA loadings (Theorem 3).

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